Mmesoma Nwachukwu

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PROFILE

AI/ML Engineer 2026 graduate with a First Class from Coventry University. Built machine learning systems at various companies and deployed production models in a startup environment. Experienced in computer vision and LLM systems, including RAG pipelines and scalable backend infrastructure. Recipient of 5 scholarships and awards across a 400-person cohort and nationwide competitions.

EDUCATION

BSc Computer Science Coventry University *May 2026*

First Class Honours Relevant modules: AI, Data Science, Advanced Algorithms, Software Engineering, Theory of Computation

SCHOLARSHIPS AND AWARDS

Year in Industry Student of the Year *National Highways, 2025*
1 of 10 selected from 400 placement students nationwide

AWS Machine Learning Scholarship *Amazon Web Services, 2023*
Top 2% of 50,000+ DeepRacer applicants

Santander Universities Scholarship *Santander Bank, 2023*
1 of 100 recipients selected nationwide

Intern of the Year *Paradigm Finance, 2021*

Best Foundation Student *16 Plus Schools, 2021*
Top student out of 50 in the cohort

PROFESSIONAL EXPERIENCE

Founding Engineer Intern Ventureship *August– September 2026*

- Built an end-to-end agentic pipeline for an AI hiring platform using the Anthropic SDK, replacing a fully manual, founder-run process with one automated pipeline run.
- Designed multi-stage agents with self-checking schemas, deterministic decision logic, and bias audits at key decision points, fixing bugs surfaced across 3 rounds of production test runs.
- Contributed to core engineering decisions across the platform, including the shared data schema and write interface with the application layer, reviewed weekly with the technical advisor.

Software Engineer (Volunteering) Samaritan's Purse *July – August 2026*

- Designed and built a digital ordering and stock management system for a £10,000 charity procurement project supplying discounted shoebox gifts to 50 community groups
- Built the entire system solo, including an online order form, automated payment and confirmation flow, and a live stock dashboard with low stock alerts.
- Replaced a manual, spreadsheet-based process with an automated pipeline to cut admin time and reduce order errors

AI Engineer (Contractor) Projekt Rising *Dec 2025 - Mar 2026*

- Analysed client workflows across several SMEs to identify automation gaps, then designed and delivered AI integrations that automated several business processes, cutting roughly 600 hours of manual work per month across the client operations.
- Built and monitored production APIs connecting AI models to client data sources, tracking accuracy, latency, and user outcomes across deployments.

Software Engineer Intern SalesAPE.ai *Dec 2025 - Mar 2026*

- Designed and built 3 MVP AI prototypes as part of a fast-moving hacksquad, delivering demos to leadership that led to 2 of the 3 projects getting greenlit for further development.
- Designed evaluation pipelines to measure model reliability and document experiment results, moving prototypes closer to production quality.

Data Science Year in Industry National Highways *Jul 2024 - Jul 2025*

- Built a weather classification and removal model using CNNs achieving 98% classification accuracy on a dataset of 10,000 CCTV images, improving the reliability of road monitoring systems.
- Developed a RAG knowledge graph over thousands of interconnected engineering documents, enabling structured insight extraction to support infrastructure asset management.
- Built a computer vision segmentation model on satellite imagery to identify and classify roadside habitats, improving environmental workflows in infrastructure planning.

Founder FaceRegg *May 2023 - Aug 2023*

- Founded and built an image verification SaaS using CNNs for facial and image matching, acquiring 2 paying clients and generating £5k in revenue through automated identity validation.
- Designed model training and inference pipelines using TensorFlow and OpenCV, optimising accuracy and response latency for real-world inputs.

Research Intern Imperial College London *Jun 2023 - Jul 2023*

- Collaborated in a cross-functional team prototyping data-driven technologies addressing climate resilience, working with researchers and industry mentors to assess impact and feasibility.

Software Engineering Intern Paradigm Finance *Sept 2021 - Dec 2021*

- Built internal tools for credit risk analysis, generating client portfolio reports and maintaining web applications while adhering to financial data security standards.

PERSONAL PROJECTS

Sentiel1

- Built a dual-branch ResNet50 system identifying vehicle make, model, and year from images in 46ms, achieving 81.71% hierarchical accuracy using ordinal regression, Kendall uncertainty weighting, and temperature scaling calibration.
- Deployed a Flask REST API with real-time Euro NCAP database lookup via fuzzy matching, producing court-admissible forensic JSON outputs for accident reconstruction pipelines.

ExcelPoint

- Built a full-stack RAG learning platform using Django, React/TypeScript, PostgreSQL, and LangChain with semantic search and context-aware chat over uploaded educational documents using vector embeddings and sentence transformers.
- Architected async processing pipelines with Celery and Redis for document analysis, quiz generation, and adaptive learning workflows, integrated with AWS S3 and JWT-authenticated APIs.

CUDA Transformer Engine from scratch

- Built a transformer from scratch using **CUDA kernels** and a custom autograd engine, implementing multi-head causal attention, matrix multiplication, softmax, and backpropagation without PyTorch or TensorFlow for the core maths.
- Trained the same transformer architecture for language modelling and next-movie recommendation on 1M MovieLens interactions, then **deployed** the recommender as a live web application with a production inference API. Optimised tiled attention to achieve 25.7x faster execution at sequence length 1024 while avoiding 128 MB of GPU memory allocation.

SKILLS

ML / AI: PyTorch, TensorFlow, LangChain, Hugging Face, OpenCV, Scikit-learn, RAG, Knowledge Graphs, Reinforcement Learning, CUDA

Backend: Python, Django, Flask, Node.js, PostgreSQL, MongoDB, Redis, Celery, REST APIs, SQL

Frontend: React, TypeScript, JavaScript, HTML/CSS

Cloud / Infra: AWS S3, Databricks, Docker, Git, Jupyter, VS Code

Languages: Python, C++, JavaScript, TypeScript, R, SQL